

2010.14265 — formula report

369 inline formulas (section omitted; all rendered), 16 display equations, 0 tables, 6 unrecovered image regions.

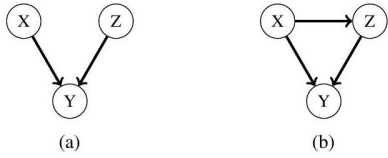
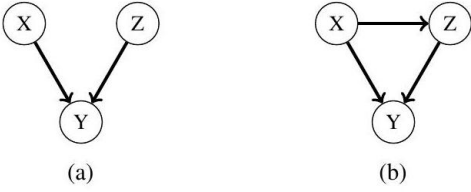
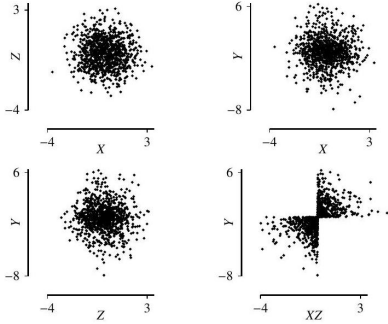
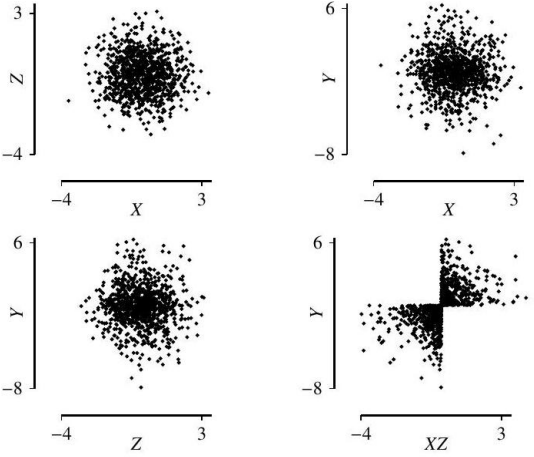
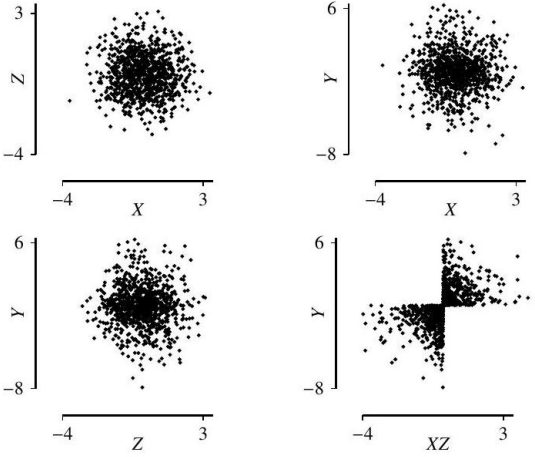
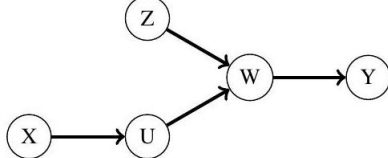
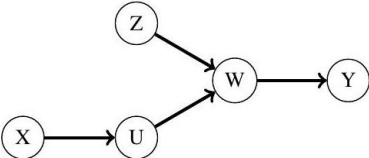
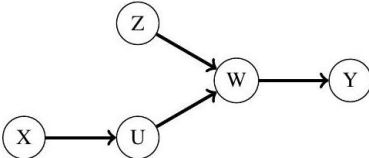
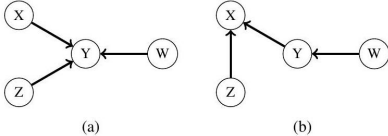
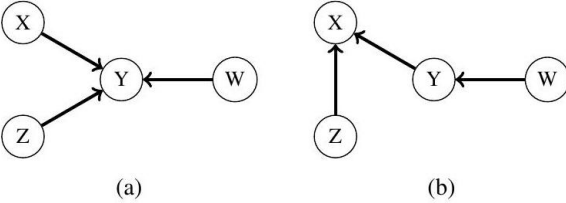
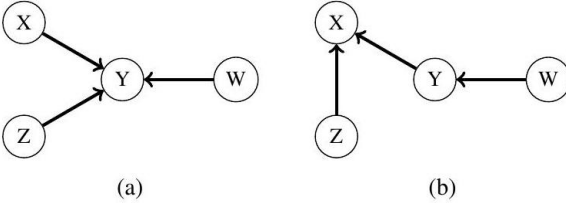
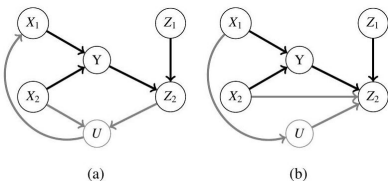
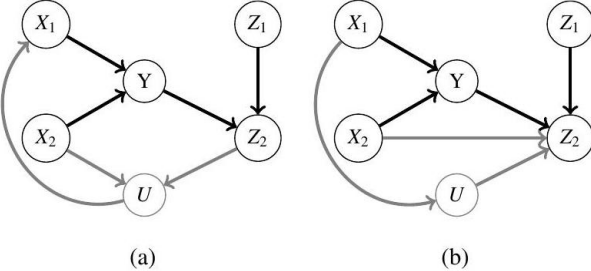
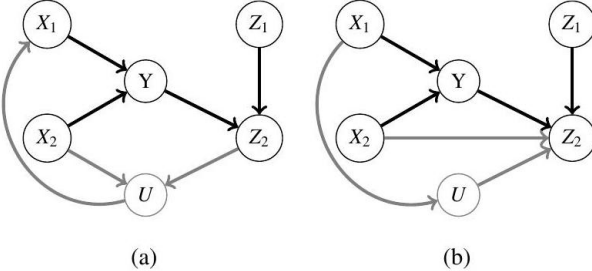
Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2010.14265_E00012	012	■ 0.492 ● U +117	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(2): Z \perp_P W \mid \emptyset \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): Z \perp_P Y \mid \{W\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): Z \not\lhd_P X \mid \{W, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): W \not\lhd_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): W \not\lhd_P X \mid \{Z, Y\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\lhd_P X \mid \{Z, W\} \end{aligned}$	(not rendered)	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(2): Z \perp_P W \mid \emptyset \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): Z \perp_P Y \mid \{W\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): Z \not\lhd_P X \mid \{W, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): W \not\lhd_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): W \not\lhd_P X \mid \{Z, Y\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\lhd_P X \mid \{Z, W\} \end{aligned}$
2010.14265_E00009	011	■ 0.578 ● U +51	$\begin{aligned} & \pi(1) \rightarrow \pi(4): X \not\perp_P Y \mid \{Z, W\} \text{ \& } \pi(2) \rightarrow \pi(4): \\ & Z \not\perp_P Y \mid \{X, W\} \text{ \& } \pi(3) \rightarrow \pi(4): \\ & W \perp_P Y \mid \{X, Z\} \end{aligned}$	(not rendered)	$\begin{aligned} & \pi(1) \rightarrow \pi(4): X \not\lhd_P Y \mid \{Z, W\} \\ & \pi(2) \rightarrow \pi(4): Z \not\lhd_P Y \mid \{X, W\} \\ & \pi(3) \rightarrow \pi(4): W \not\lhd_P Y \mid \{X, Z\} \end{aligned}$
2010.14265_E00008	011	■ 0.675 ● U +34	$X_{\{\pi(j)\} \not\perp_P X_{\{\pi(k)\} \mid \left\{X_{\{\pi(1)\}, X_{\{\pi(2)\}, \ldots, X_{\{\pi(k-1)\}}\right\} \backslash \left\{X_{\{\pi(j)\}}\right\},}$	(not rendered)	$X_{\pi(j)} \not\lhd_P X_{\pi(k)} \mid \{X_{\pi(1)}, X_{\pi(2)}, \ldots, X_{\pi(k-1)}\} \backslash \{X_{\pi(j)}\} \text{ ,}$
2010.14265_E00013	013	■ 0.706 ● U +2	$Z \perp_{\mathcal{G}} \{X, Y\} \mid \operatorname{Pa}(Z) \text{ .}$	(not rendered)	$Z \perp_G \{X, Y\} \mid \operatorname{Pa}(Z) \text{ .}$
2010.14265_E00001	004	■ 0.823 ● U +23	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y, Z\}: A \not\perp_P B \mid \boldsymbol{S} \cup \{X, Y, Z\} \backslash \{A, B\} \text{ ,}$	(not rendered)	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y, Z\}: A \not\lhd_P B \mid \boldsymbol{S} \cup \{X, Y, Z\} \backslash \{A, B\} \text{ ,}$
2010.14265_E00006	011	■ 0.874 ● N +0	$\begin{aligned} & P(X=1, \text{ \& } W=1, Y=1, Z=1)=\frac{pq}{4}, \text{ \& } \\ & \text{ \& } P(X=1, Y=1, Z=1)=\frac{q}{4}, \text{ \& } P(W=1, Y=1, Z=1)= \\ & \frac{p}{4} \text{ . } \end{aligned}$	$\begin{aligned} P(X=1, W=1, Y=1, Z=1) &= \frac{pq}{4}, \\ P(X=1, Y=1, Z=1) &= \frac{q}{4}, \\ P(W=1, Y=1, Z=1) &= \frac{p}{4}. \end{aligned}$	$\begin{aligned} P(X=1, W=1, Y=1, Z=1) &= \frac{pq}{4} \text{ ,} \\ P(X=1, Y=1, Z=1) &= \frac{q}{4} \text{ ,} \\ P(W=1, Y=1, Z=1) &= \frac{p}{4} \text{ .} \end{aligned}$
2010.14265_E00015	013	■ 0.960 ● U +14	$X \perp_{\mathcal{G}} Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup (\boldsymbol{Z} \backslash \{Z\}) \text{ ,}$	(not rendered)	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup (\boldsymbol{Z} \backslash \{Z\}) \text{ ,}$
2010.14265_E00014 (1)	013	■ 0.980 ● U +14	$X \perp_{\mathcal{G}} Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup (\boldsymbol{Z} \backslash \{Z\}) \text{ ,}$	(not rendered)	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup (\boldsymbol{Z} \backslash \{Z\}) \text{ ,}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2010.14265_E00011	012	■ 0.998 ● U +99	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): X \not\perp_P Y \mid \{Z\} \quad \& \quad \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): Z \not\perp_P Y \mid \{X\} \\ & Y \mid \{X\} \quad \& \quad \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P W \mid \{X, Z\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): X \not\perp_P W \mid \{Z, Y\} \quad \& \quad \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): Z \not\perp_P W \mid \{X, Y\} \end{aligned}$	(not rendered)	$\begin{aligned} & \pi'(1) \rightarrow \pi'(3): X \not\perp_P Y \mid \{Z\} \\ & \pi'(2) \rightarrow \pi'(3): Z \not\perp_P Y \mid \{X\} \\ & \pi'(3) \rightarrow \pi'(4): Y \not\perp_P W \mid \{X, Z\} \\ & \pi'(1) \rightarrow \pi'(4): X \not\perp_P W \mid \{Z, Y\} \\ & \pi'(2) \rightarrow \pi'(4): Z \not\perp_P W \mid \{X, Y\} \end{aligned}$
2010.14265_E00010	012	■ 1.000 ● U +39	$\begin{aligned} & \pi(1) \rightarrow \pi(2): X \perp_P Z \mid \emptyset \\ & \pi(1) \rightarrow \pi(3): X \perp_P W \mid Z \\ & \pi(2) \rightarrow \pi(3): Z \perp_P W \mid X . \end{aligned}$	(not rendered)	$\begin{aligned} & \pi(1) \rightarrow \pi(2): X \perp_P Z \mid \emptyset \\ & \pi(1) \rightarrow \pi(3): X \perp_P W \mid Z \\ & \pi(2) \rightarrow \pi(3): Z \perp_P W \mid X . \end{aligned}$
2010.14265_E00016	013	■ 1.000 ● U +14	$X \perp_P Z \mid \boldsymbol{S} \cup \{Y\} \cup (X \backslash \{X\}) \cup (Z \backslash \{Z\}) .$	(not rendered)	$X \perp_P Z \mid S \cup \{Y\} \cup (X \backslash \{X\}) \cup (Z \backslash \{Z\}) .$
2010.14265_E00002	006	■ 1.000 ● K +0	$Y := ((X \oplus Z) \wedge W) \oplus E,$	$Y := ((X \oplus Z) \wedge W) \oplus E,$	$Y := ((X \oplus Z) \wedge W) \oplus E ,$
2010.14265_E00003	011	■ 1.000 ● K +0	$Y := ((X \oplus Z) \wedge W) \oplus E .$	$Y := ((X \oplus Z) \wedge W) \oplus E.$	$Y := ((X \oplus Z) \wedge W) \oplus E .$
2010.14265_E00004	011	■ 1.000 ● W +0	$\begin{aligned} & P(X=0, Z=1, Y=1) \quad \& \quad = P(X=0, Z=1) \cdot P(Y=1) \\ & P(Y=1) \quad \& \quad \Leftrightarrow p+q-2pq = q + \frac{p}{2} - pq \\ & \quad \& \quad \Leftrightarrow p-pq = \frac{p}{2} . \end{aligned}$	$\begin{aligned} & P(X=0, Z=1, Y=1) = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p+q-2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p-pq = \frac{p}{2} . \end{aligned}$	$\begin{aligned} & P(X=0, Z=1, Y=1) = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p+q-2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p-pq = \frac{p}{2} . \end{aligned}$
2010.14265_E00005	011	■ 1.000 ● N +0	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)} .$	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)} .$	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)} .$
2010.14265_E00007	011	■ 1.000 ● N +0	$p \, q = \frac{p \, q}{2 P(Y=1)} .$	$pq = \frac{pq}{2 P(Y=1)} .$	$pq = \frac{pq}{2 P(Y=1)} .$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
2010.14265_OIA_0001	001	— (dup)	58d90493-1e15-41f0-8b44-298986e56d74-01_256_632_859_1156.jpg			
2010.14265_OIA_0002	003	— (dup)	58d90493-1e15-41f0-8b44-298986e56d74-03_602_722_221_1126.jpg			
2010.14265_OIA_0003	005	— (dup)	58d90493-1e15-41f0-8b44-298986e56d74-05_217_493_213_348.jpg			
2010.14265_OIA_0004	006	— (dup)	58d90493-1e15-41f0-8b44-298986e56d74-06_276_758_210_1093.jpg			
2010.14265_OIA_0005	007	— (dup)	58d90493-1e15-41f0-8b44-298986e56d74-07_367_792_210_1075.jpg			
2010.14265_OIA_0006	009	— (dup)	no image for this region (5 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)